

CSE114 Midterm 1 - Practice Problems

1) What is the output of the following statement?

```
System.out.println((2.0 + 3) + 3);
```

- a) 5.03
- b) 8.0
- c) 8
- d) 53
- e) 5.3

2) What is the output of the following code?

```
for (int c = 'A'; c < 'F'; c++){  
    System.out.print(c + " ");  
}
```

- a) A B C D E
- b) 65 66 67 68 69
- c) There is a compiler error.
- d) None of the above.

3) What is the output of the following code?

```
double x = (int)2.0;  
double f = (x / (int)3.0);  
System.out.println(f);
```

- a) 0
- b) 0.0
- c) 0.66
- d) 0.6666666666666666
- e) 1

4) What is returned by the following method?

```
public static String oneOfTwo(){  
    int x = (6/4 == 1)?2:3;
```

```
        return x;
    }
```

- a) 2
- b) 3
- c) "2"
- d) "3"
- e) Compiler error.

5) What is the output of the following program?

```
public class Test{
    static boolean a = false;
    static boolean b = !a;
    static boolean c = (25 % 3 == 0 || 2 != 3);

    public static void main(String[] args){
        if (a && !a){
            if(c)
                System.out.println("32");
            else System.out.println("31");
        }
        else if(c || b && a){
            if(c)
                System.out.println("30");
            else System.out.println("29");
        }
        else System.out.println("28");
    }
}
```

- a) 32
- b) 31
- c) 30
- d) 29
- e) 28

6) Write a static public method called `encrypt`, that takes one `String` as an argument and returns a `String`. The method encrypts a word, by replacing each letter with the character two places ahead of it in the alphabet. The last two letters of the alphabet wrap around to the front (i.e. 'y' = 'a', 'z' = 'b'). Assume that all input consists solely of lowercase letters, and that only one word is input.

Sample output:

```
System.out.println(encrypt("abcdefxyz"));  
cdefghzab  
  
System.out.println(encrypt("sampletext"));  
ucornvgzv
```

7) Write a static public method called reverse, that accepts one String as an argument, returns nothing, and prints the reverse of the input.

Sample output:

```
reverse("The quick brown fox jumps over the lazy dog.");  
.god yzal eht revo spmuj xof nworb kciuq ehT
```

8) Create a class, called `TriangleNumbers`, that has a static public method called `genNum` that takes one non-negative integer argument `n`, and returns the n^{th} triangle number, an integer. The n^{th} triangle number is defined as the sum of numbers 1 through `n`, inclusive. For example:

First triangle number = 1

Third triangle number = $1 + 2 + 3 = 6$

Ninth triangle number = $1 + 2 + 3 + \dots + 8 + 9 = 45$

The program should prompt the user to enter an integer `x` (you can safely assume it is greater than 0), and then print the first `x` triangle numbers, with each i^{th} line holding the i^{th} triangle number amount of values, separated by commas (except for the last number in every line). So the first line would contain the first triangle number, the second line would contain the next three, the third line would contain the next six, the fourth line would contain the next ten, and so forth.

Sample input/output:

User enters 4, program outputs:

1

3, 6, 10

User enters 22, program outputs:

1

3, 6, 10

15, 21, 28, 36, 45, 55

66, 78, 91, 105, 120, 136, 153, 171, 190, 210

231, 253

9) Write a public static method `printHourglass` that accepts one integer argument, returns nothing, and prints an hourglass pattern.

Sample output:

`printHourglass(3)`

*

`printHourglass(6)`

* *

**

* *

`printHourglass(14)`

* *

* *

* *

* *

* *

```

      * *
    *   *
  *     *
*       *
*     *
*   *
* *
* * * * *

```

10) Problem taken from projecteuler.net (#1)

If we list all the natural numbers below 10 that are multiples of 3 or 5, we get 3, 5, 6 and 9. The sum of these multiples is 23. Find the sum of all the multiples of 3 or 5 below 1000.